

Urvish Shah

Buffalo, NY, USA (*open to relocation*)

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Professional Summary

Robotics engineer with hands-on experience building, deploying, and debugging physical robotic systems on real hardware. Strong background in ROS2-based control, slam, mechatronics, and hardware–software integration, with extensive real-world testing beyond simulation. Experienced in system bring-up, failure analysis, and making robots work reliably under real operating constraints.

Technical Skills

Robotics & Systems: ROS, ROS2, SLAM, system integration, mechatronics, real-world testing

Hardware & Embedded: PCB design (Altium), ARM/AVR/ESP32, sensors, actuators

Programming: C++, Python, Embedded C

Algorithms: Bayesian inference, Gaussian Processes

CAD & Mechanical Design: SolidWorks (part modeling, assemblies, drawings)

Prototyping: 3D printing, mechanical assembly, experimental validation

Education

Master of Science in Engineering Science (Robotics)

Aug 2024 – Dec 2025

University at Buffalo, State University of New York

GPA: 3.63 / 4.0

Thesis: *Decentralized Multi-Robot Collaborative Transport and Source Localization*

Bachelor of Engineering in Instrumentation and Control Engineering

Jun 2019 – Aug 2023

Government Engineering College, Gandhinagar, India

Engineering & Research Experience

Thesis: *Decentralized Multi-Robot Collaborative Transport and Source Localization* Aug 2024 – Dec 2025

ADAMS Lab, University at Buffalo

- Designed and deployed a decentralized collaborative transport system using four physical TurtleBot3 robots.
- Enabled coordinated manipulation of objects $\sim 4\times$ larger than a single robot footprint and $\sim 2\times$ heavier than individual robot mass.
- Achieved stable transport on ramps up to 25° inclination through coordinated force distribution and failure handling.
- Built ROS2-based distributed control, coordination, and monitoring pipelines validated on real hardware.
- Implemented geometric, rule-based role assignment (push, support, stabilize) using relative robot–object poses.
- Integrated Bayesian swarm-based source localization using Gaussian Processes and LoRa RSSI sensing.
- Performed extensive sim-to-real validation, debugging hardware failures, communication dropouts, and control edge cases.

Research Intern → Project Assistant, Smart Manufacturing Systems

Jan 2023 – Apr 2024

Indian Institute of Technology Gandhinagar

- Designed and built a pellet-based (filamentless) 3D printing system for recycled plastics.
- Developed a custom pellet-fed extruder and integrated on-machine shredding for waste plastic reuse.
- Designed control electronics and firmware for extrusion, thermal regulation, and material flow.
- Conducted experimental benchmarking, failure analysis, and repeatability studies on physical systems.
- Work contributed to peer-reviewed journal publications in additive manufacturing.

Hardware Design Engineer (Intern)

Jun 2022 – Aug 2022

OoB Services, Ahmedabad

- Designed and validated multi-layer PCBs using Altium Designer.
- Integrated microcontrollers, sensors, wireless modules, and power electronics.
- Performed board bring-up, debugging, and system-level validation.

Robotics Hardware Engineer

Aug 2020 – May 2023

GTU ABU Robocon Team, Ahmedabad, India

- Developed embedded control systems for competition robots using Arduino and STM32 microcontrollers.

- Designed and tested PCBs for motor drivers and actuator control; integrated IMUs, encoders, motors, distance sensors, and pneumatic mechanisms.
- Implemented closed-loop motor control and sensor-based actuation for a robotic arrow-throwing task.
- Assisted with system integration, testing, competition deployment, and mentoring junior members.
- Co-inventor on Indian patent IN202321008858.

Publications

- Shah, U., et al. (2025). *Sustainable Recycling of ABS: Thermophysical Characterisation in Filament vs. Granules-Based 3D Printing*. Rapid Prototyping Journal.
- Shah, U., et al. (2025). *Multi-Sensor Deep Learning Framework for Detection of Nozzle Clogging in Pellet-Based 3D Printing*. Progress in Additive Manufacturing.

Selected Projects

Scarecrow 2.0: Agricultural Innovation Project

- Integrated IoT devices, PIR sensors, GSM modules, and rain sensors for pest deterrence and animal detection in agricultural settings.
- Implemented detection and alerting logic; achieved $\sim 70\%$ detection accuracy during field testing.
- Worked on renewable energy integration to support low-power, outdoor operation.
- Pitched the project at a college Ideathon hosted by the Entrepreneurship Cell in collaboration with IIT Bombay; received runner-up recognition.

Achievements

- Represented India at ABU Robocon 2021 (Jimo, China); placed 9th out of 21 teams from 8 countries.
- Received the Best Design Award (2022) for robot mechanical and system design.
- Co-developed a pellet-based 3D printer using waste plastic; awarded Best Design at the Vishvakarma Awards, IIT Delhi.